University of Chicago **Spring 2026**
Department of Economics **ECMA 31150**

Problem Set 2

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Please upload your answers on Canvas, with code attached. You are encouraged to collaborate, but please write up your answers independently and do not share code chunks. Use any software you like, although you learn the most by limiting the use of generative AI. If you find any errors or require clarification, please let me know right away.

Question 1

Consider a bivariate VAR(2) model for (y_t, x_t):

$$\begin{aligned} y_t &= 1 + 0.7y_{t-1} + 0.1y_{t-2} + 0.2x_{t-1} - 0.2x_{t-2} + u_{1t}, \\ x_t &= 0.6x_{t-1} + 0.3x_{t-2} + 0.2y_{t-2} + u_{2t}, \end{aligned}$$

where u_{1t} and u_{2t} are i.i.d. bivariate normally distributed, each with variance 1, and $\text{Corr}(u_{1t}, u_{2t}) = 0.4$.

i) Is the model covariance stationary?

ii) Does $\{x_t\}$ Granger-cause $\{y_t\}$? Does $\{y_t\}$ Granger-cause $\{x_t\}$?

iii) What is the MSFE of the optimal one-step-ahead forecast of y, i.e., $E[y_{t+1} \mid \{y_s, x_s\}_{s \leq t}]$? What is the MSFE of the optimal two-step-ahead forecast of y, i.e., $E[y_{t+2} \mid \{y_s, x_s\}_{s \leq t}]$?

iv) What is $E(y_t)$, $E(x_t)$, $\text{Var}(y_t)$, $\text{Var}(x_t)$, and $\text{Corr}(y_t, x_t)$?

v) Simulate a very large data set with $T = 10000$ from this VAR model. Use the simulation output to (approximately) verify your calculations in part (iv).

[Hint: some of these questions will be hard to answer with "pen and paper", so feel free to solve them numerically on the computer.]

Question 2

Consider a univariate AR(p) model

$$y_t = c + a_1 y_{t-1} + a_2 y_{t-2} + \cdots + a_p y_{t-p} + u_t, \quad u_t \sim WN(\sigma_u^2).$$

Let $z_1, \ldots, z_p$ denote the p (possibly complex) roots of the characteristic equation

$$1 = a_1 z + a_2 z^2 + \cdots + a_p z^p,$$

and assume these are all distinct. Finally, let $\tilde{\boldsymbol{A}}$ be the $p \times p$ companion matrix that re-expresses the univariate AR(p) model as a multivariate VAR(1) model.

i) Show that the eigenvalues and corresponding eigenvectors of $\tilde{\boldsymbol{A}}$ are given by

$$z_i^{-1} \quad \text{and} \quad (1, z_i, z_i^2, \ldots, z_i^{p-1})',$$

respectively, for $i = 1, \ldots, p$.

ii) How does the result in part (i) relate to the methods for checking covariance stationarity that we have discussed in lecture? Be brief.

Question 3

Go to the St. Louis Fed's FRED website and download data on real GDP (FRED code: GDPC1), the consumer price index (code: CPIAUCSL), and consumer sentiment (code: UMCSENT). Collapse the last two series to quarterly averages.[1] Then transform real GDP and the consumer price index to quarterly log growth rates. Use only data from the beginning of 1960 until (and including) the end of 2019. Consider a VAR model in the three variables (real GDP growth, CPI inflation, consumer sentiment).

i) What lag length is selected by the AIC for the VAR? Use a maximum potential lag length of 12 quarters.

Regardless of the result from part (i), we henceforth assume a VAR(1) model.

ii) Estimate the model. Is the estimated model covariance stationary?

[1] Note that UMCSENT is recorded at a quarterly frequency until 1978.

iii) Assume (for this part only) that the VAR innovations are i.i.d. Gaussian. Compute an 80% forecast interval for inflation in the first quarter of 2020. [Hint: we did not explicitly discuss how to construct VAR forecast intervals in lecture, but you should be able to adapt the arguments from the slides on AR models.]

iv) Compute 90% bootstrap confidence intervals for each of the coefficients in the equation for inflation. Report these confidence intervals alongside the point estimates in a nicely formatted table. Use at least 500 bootstrap iterations.